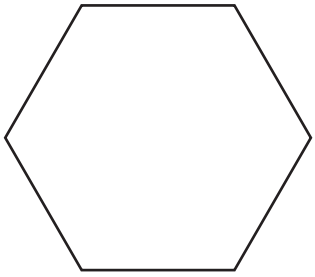
0580_s20_qp_33_Q7

7 **(a)** The diagram shows a regular polygon.



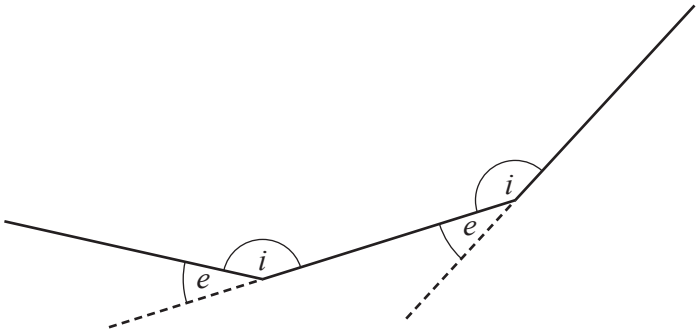
(i) Write down the mathematical name for this shape.

..... [1]

(ii) Write down the order of rotational symmetry of this shape.

..... [1]

(b) The diagram shows part of a different regular polygon.



NOT TO SCALE

e is an exterior angle.
i is an interior angle.

The ratio $e : i = 2 : 13$.

(i) Work out angle e.

..... [3]

(ii) Work out the number of sides of this regular polygon.

..... [1]

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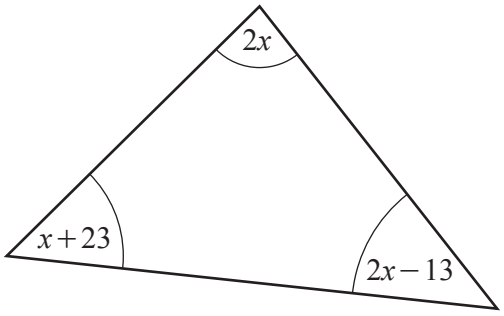
(c) **Using a straight edge and compasses only**, construct the equilateral triangle ABC.
Side AB has been drawn for you.

[2]

(d) In this part, all angles are in degrees.



NOT TO SCALE

(i) Use the information in the triangle to write down an equation in terms of x.

..... [1]

(ii) Solve this equation to find the value of x.

$x =$ ..... [3]

(iii) Work out the size of the smallest angle in the triangle.

..... [2]